

Before the Machine

From quarks to the physical requirements of fusion

We will build the machine from its purpose. First we must understand the matter we intend to change, the energy available from that change, and the obstacle between possibility and a useful reaction rate.

0.1. Inside a nucleon

A quark is one of the elementary constituents in our present theory of matter. To locate it, begin with an atom: a positively charged nucleus surrounded by negatively charged "quantum clouds" of electrons. Electrons are also elementary constituents of matter and carry charge $-e$. Chemical reactions change electron arrangements while leaving the nuclei intact. Nuclear reactions such as fission and fusion change the nuclear arrangement itself.

Inside the nucleus are *nucleons*: Protons and neutrons are baryons— composite particles made of three quarks. Quarks come in different flavors. Two types of quarks are up quarks and down quarks. A proton can be written as uud and a neutron can be written as udd , where u and d denote up and down quarks. Quarks possess both electric charge and color charge.

The electromagnetic force is described by rotations of a circle, called the $U(1)$ group, which transforms via a matrix representation of the groups linear map $\chi(\theta) = \exp(iq\theta)$ that satisfies a circles compactification: $\chi(\theta + 2\pi) = \chi(\theta)$, which quantizes charge into an integer, $q \in \mathbb{Z}$. Physics doesn't tell us why the electric charge is $+2e/3$ for an up quark and $-e/3$ for a down quark, experiments do. Given this fact, because constituent phase factors multiply (ie $e^{iq_1\theta}e^{iq_2\theta} = e^{i(q_1+q_2)\theta}$), protons have $+e$ electric charge, and neutrons have *zero net electric charge*:

$$q_p = \left(\frac{2}{3} + \frac{2}{3} - \frac{1}{3}\right)e = e, \quad q_n = \left(\frac{2}{3} - \frac{1}{3} - \frac{1}{3}\right)e = 0.$$

These are *quark labels*: they only specify net quark content. A nucleon is a quantum state that also contains gluon fields and quark–antiquark fluctuations.^[1]

What binds that state? Quarks second kind of charge, called *color*. The theory of color interactions is *quantum chromodynamics* (QCD), and it describes the strong nuclear force— one of the four fundamental forces of nature. Gluons mediate the strong force and possess color charge, leading to self-interactions amongst gluons. This is unlike photons, neutrally charged electromagnetic force carriers that have no direct electric-charge self-interaction, unlike gluons with color charge.

Color is not a single number q to be summed into a different state of color charge — it is a basis vector of $\mathbb{C}^3$, such as $(1, 0, 0)$, to be rotated via matrices whose inverse equal it's conjugate transpose and whose determinant equals to 1, eg elements of $SU(3)$. A state is *color neutral* when it is left invariant by every such rotation: for baryons this is achieved by carrying exactly one quark of each color, combined in the unique totally antisymmetric combination, $\epsilon_{ijk}q^iq^jq^k$, that $SU(3)$ leaves unchanged. This explains why three quarks can form a color-neutral state, whereas two or four quarks alone cannot: three colors demand three quarks to form this invariant, fully analogous to how a determinant vanishes unless you antisymmetrize over as many vectors as there are dimensions.^[2]

Most nucleon mass is not the sum of the small up- and down-quark mass parameters. It comes from the energy of the interacting quark–gluon system. Lattice QCD, which approximates the fields on a spacetime grid and controls extrapolations, has reproduced the light-hadron mass spectrum. Most ordinary matter's mass is in nucleons, and most nucleon mass is dynamical QCD energy: these are two distinct statements. Neither assigns a universal percentage of matter's mass to the geometric flux-tube

picture.^[3]

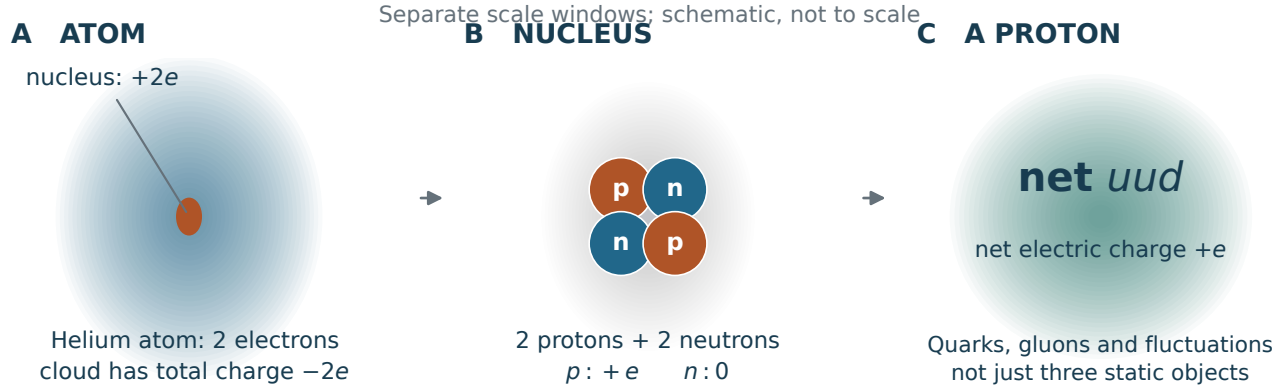


Figure 1: Inside a helium atom, then a proton. The separate scale windows show constituent bookkeeping, not classical particle trajectories or a calculated density. Colored regions are schematic; the proton’s *uud* label specifies net valence content, not its entire quantum state. Original diagram; physical interpretation follows sources 1–3.

0.2. Confinement, and the boundary of the explanation

QCD has an unusual scale dependence. At sufficiently high momentum transfer, which probes short distances, its effective coupling becomes small. This is *asymptotic freedom*. It does not mean that a quark can leave a proton for free: short-distance scattering and separating a quark into an isolated outgoing particle are different questions. At longer distances, the weak-coupling expansion ceases to be reliable. Extrapolating that expansion is not a proof of confinement.^[4]

Color confinement is the absence of isolated colored particles in the observable outgoing states of ordinary low-energy experiments. A useful picture comes from slowly separating a very heavy quark and antiquark. Over an intermediate range, their color-field energy concentrates in an elongated region, or *flux tube*. In its approximately linear regime,

$$V(r) \simeq V_0 + \mathcal{T}_s r, \quad F_r = -\frac{dV}{dr} \simeq -\mathcal{T}_s.$$

Here r is separation, V_0 an irrelevant reference energy, and $\mathcal{T}_s$ the string tension, with units of energy per length, hence force. More separation requires more work; the attraction need not grow without bound. With dynamical quarks, sufficient energy can instead form an additional quark–antiquark pair: the string breaks into color-neutral hadrons, not free quarks. Lattice calculations resolve this process. This static-source picture is not a literal photograph of three rigid strings inside a moving proton.^[5]

An open mathematical problem, not an absence of physics The Yang–Mills existence and mass-gap problem asks for a rigorous construction of the four-dimensional quantum theory and a strictly positive energy gap above its vacuum. Writing H for its energy operator and setting vacuum energy to zero, the gap condition is

$$\text{spec}(H) \cap (0, \Delta) = \emptyset \quad \text{for some } \Delta > 0.$$

No physical excitation can have an energy strictly between zero and Δ . The official problem concerns pure Yang–Mills theory; confinement in QCD with quarks is a related but distinct question. Neither follows just by quoting asymptotic freedom.^[6]

Experiments, numerical evidence, useful approximations, and an axiomatic proof are different kinds of knowledge. The missing proof should be acknowledged precisely; it does not make nuclear energy release mysterious or permit an arbitrary mechanism to replace tested nuclear physics.

There is an analogue outside nuclear physics: in the *Kondo effect*, the effective interaction between a magnetic impurity and a metal’s electrons is weak at high energies but becomes strong at low energies, where collective screening develops. This is an analogy in scale-dependent coupling, not evidence that ordinary magnetic fields confine quarks or that a fusion reactor contains QCD flux tubes.^[7]

A Nearby sources



external heavy sources

B Work stretches the tube

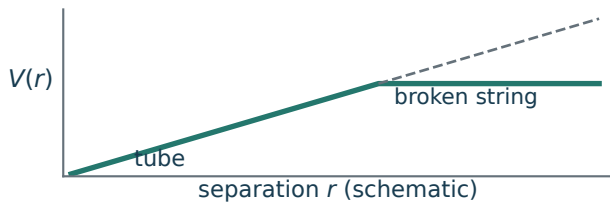


approximately linear energy cost

C A light pair forms



two color-neutral mesons



Solid: lower-energy branch
Dashed: unbroken-string branch

Qualitative crossover, no physical scale.
Not a picture of a proton.
No isolated colored products.

Figure 2: Static heavy sources $Q, \bar{Q}$ and creation of a light $q, \bar{q}$ pair. The final groups are color neutral; dashed ovals group constituents rather than marking physical walls. The qualitative potential sketch omits the smooth mixing near string breaking and has no physical scale. Original schematic based on source 5; not a proton model.

0.3. From confined quarks to interacting nuclei

If each nucleon is color neutral, how can nucleons bind to one another? Neutrality does not mean that a composite object has no interactions. Even electrically neutral atoms can interact through their internal charged constituents. That comparison illustrates the logical possibility of a residual force; it does not identify the nuclear force with an electromagnetic one.

At nuclear energies, a practical description replaces explicit quark and gluon variables by nucleons and, in one common formulation, pions. Pions are light quark–antiquark composite particles. Their virtual exchange describes the long-range part of the effective nuclear interaction; shorter-range terms and forces involving three or more nucleons are also needed. These are parts of an effective field theory, organized by QCD’s symmetries and a low-energy expansion, with parameters constrained by data. Its accuracy has a regime and an uncertainty.^[8]

Residual interaction does not mean leaking energy. An isolated stable proton is not steadily losing its mass into nearby space. Instead, the energy of a system of interacting nucleons depends on their joint quantum state. Some arrangements are bound: separating them into free nucleons requires energy. When a reaction reaches a lower-rest-energy arrangement, conservation of energy requires the difference to appear elsewhere, for example as kinetic energy. That is the connection we need from QCD to fusion.

1. Reading the reaction before using it

Let Z count protons and N count neutrons; $A = Z + N$ counts nucleons. The symbol A_ZX specifies a nucleus. Changing Z changes the element X ; changing N at fixed Z gives a different isotope. For our fuel and products,

Nucleus	Symbol	Constituents	Electric charge
Deuterium (deuteron)	$D = {}^2_1\text{H}$	$1p + 1n$	$+e$
Tritium (triton)	$T = {}^3_1\text{H}$	$1p + 2n$	$+e$
Helium-4 (alpha)	$\alpha = {}^4_2\text{He}$	$2p + 2n$	$+2e$
Free neutron	n	unbound neutron	0

The principal D–T fusion channel is



Its bookkeeping is $2 + 3 = 4 + 1$ nucleons and $1 + 1 = 2 + 0$ units of charge. The neutron on the right is not inside the helium nucleus. Energy, momentum, and angular momentum must also balance, but these conditions do not force an encounter to fuse: the nuclei can scatter without changing species.^[9]

The Sun’s proton–proton route includes $p + p \rightarrow D + e^+ + \nu_e$, where the weak interaction converts a proton into a neutron during formation of deuterium; e^+ is a positron and ν_e an electron neutrino. Our D–T reaction already has neutron-containing fuel and needs no such weak conversion.^[10]

2.1. The energy is a difference between arrangements

We take rest-energy equivalence, $E_{\text{rest}} = mc^2$, as an established physical input, with c the speed of light. Define the positive nuclear binding energy by

$$B(Z, N) = [Zm_p + Nm_n - m(Z, N)]c^2. \quad (2)$$

Here $m(Z, N)$ is the mass of the bare nucleus, not the neutral atom. The definition compares that nucleus at rest with the same numbers of isolated protons and neutrons at rest. Its rearrangement is $m(Z, N)c^2 = Zm_pc^2 + Nm_nc^2 - B(Z, N)$.

The energy made available by one D–T reaction, called its Q value, is the initial rest energy minus the final rest energy. Substituting the three binding energies shows exactly what cancels:

$$\begin{aligned}
 Q &= (m_D + m_T - m_\alpha - m_n)c^2 \\
 &= [(m_p + m_n)c^2 - B_D] + [(m_p + 2m_n)c^2 - B_T] \\
 &\quad - [(2m_p + 2m_n)c^2 - B_\alpha] - m_nc^2 \\
 &= B_\alpha - B_D - B_T.
 \end{aligned} \tag{3}$$

The nucleons' large rest energies cancel. We are not extracting all the QCD energy that constitutes their masses. Evaluated mass data give, rounded to 0.001 MeV, $B_D \simeq 2.225$, $B_T \simeq 8.482$, and $B_\alpha \simeq 28.296$ MeV. Thus

$$Q \simeq 28.296 - 2.225 - 8.482 = 17.589 \text{ MeV} \simeq 17.6 \text{ MeV}.^{[11]}$$

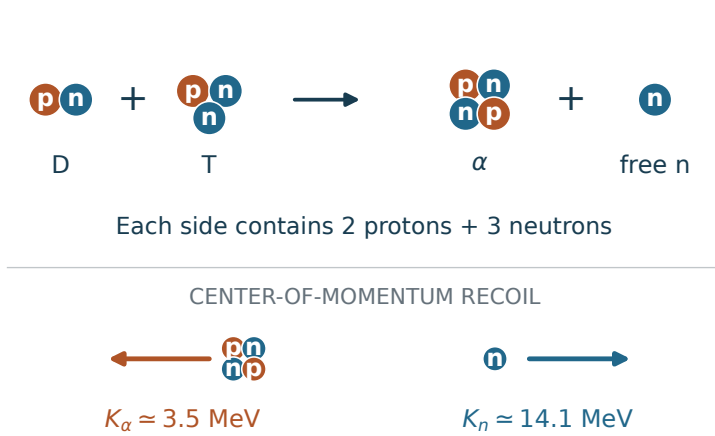
An electronvolt (eV) is the energy gained by charge e across one volt; $1 \text{ MeV} = 10^6 \text{ eV}$ and $1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}$.^[12] The rounded binding values are measured inputs, not a QCD calculation performed here.

Where does the released energy go? In the center-of-momentum frame, total momentum is zero. Therefore $\mathbf{p}_\alpha + \mathbf{p}_n = 0$, so the two momentum magnitudes equal some p . For a transparent leading estimate take nonrelativistic kinetic energies and neglect incident kinetic energy compared with Q :

$$K_\alpha = \frac{p^2}{2m_\alpha}, \quad K_n = \frac{p^2}{2m_n}, \quad \frac{K_n}{K_\alpha} = \frac{m_\alpha}{m_n} \simeq 4, \quad K_n + K_\alpha \simeq Q.$$

Consequently $5K_\alpha \simeq Q$, giving $K_\alpha \simeq 3.5 \text{ MeV}$ and $K_n \simeq 14.1 \text{ MeV}$. Precise values use relativistic two-body kinematics and the incident energy; laboratory energies also depend on center-of-momentum motion.^[13]

A Same nucleons, different arrangement



B Rest-energy difference

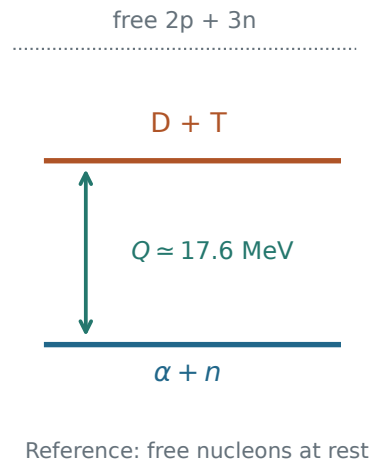


Figure 3: The same $2p+3n$ inventory is rearranged. The two final recoil arrows have equal momentum magnitude, not equal kinetic energy; energies use the leading nonrelativistic estimate in the text. Rest energies are measured relative to separated nucleons. Nucleon circles encode counts, not exact positions or nuclear shapes. Original diagram using sources 11 and 13 and Eq. (3).

2.2. An energetic downhill route can still have a barrier

The wider pattern is visible by plotting binding energy *per nucleon*, B/A , against nucleon count A . It rises broadly from light nuclei to the iron–nickel region and decreases toward very heavy nuclei. Many light-nucleus fusion reactions and heavy-nucleus fission reactions therefore release energy: their products have greater *total* binding energy than their reactants. Neither statement says that every possible fusion or fission channel releases energy, or that matter automatically evolves to iron.^[9]

Why a peak? The effective nuclear attraction is short ranged: a nucleon mostly benefits from nearby neighbours. Small nuclei have a large surface fraction, so many nucleons have fewer neighbours; larger nuclei initially gain more binding per particle. But electric repulsion acts between every pair of protons, making large charged nuclei increasingly costly. This competition gives the broad trend. Quantum shell structure, proton–neutron imbalance, and pairing determine the detailed curve; this argument alone does not calculate the peak.^[9]

Iron is a useful landmark, not a unique universal definition of stability. ^{62}Ni has the largest measured B/A , about 8.795 MeV; ^{56}Fe is nearby at about 8.790 MeV. Stability against a particular decay still depends on its possible products and conserved quantities.^[11]

Being downhill is not the same as being accessible. A deuteron and triton both carry $+e$. Taking their separation r outside the nuclear-overlap region, Coulomb’s law gives a repulsive force $F_r = e^2/(4\pi\epsilon_0 r^2)$. With $U(\infty) = 0$ and $F_r = -dU/dr$, integration gives

$$U(r) = \int_r^\infty \frac{e^2}{4\pi\epsilon_0 s^2} ds = \frac{e^2}{4\pi\epsilon_0 r}. \quad (4)$$

At $r = 5 \text{ fm}$, where $1 \text{ fm} = 10^{-15} \text{ m}$, this point-charge estimate is $1.440/5 \simeq 0.288 \text{ MeV}$. It illustrates a barrier scale, not a sharp nuclear surface or an exact fusion threshold.

Quantum mechanics permits a wave to penetrate a region forbidden to a classical trajectory. To see why, use a *one-dimensional constant-barrier model* with height $U_0 > E$ and effective mass μ . The stationary Schrödinger equation, an imported quantum law, becomes

$$-\frac{\hbar^2}{2\mu}\psi'' + U_0\psi = E\psi \quad \Rightarrow \quad \psi'' = \chi^2\psi, \quad \chi = \frac{\sqrt{2\mu(U_0 - E)}}{\hbar}.$$

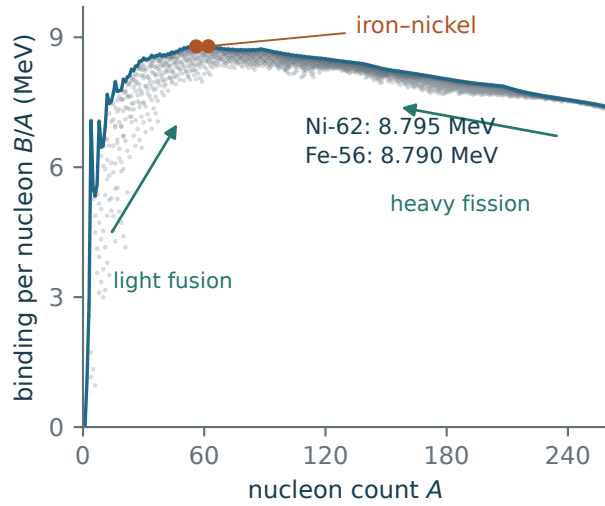
Here ψ is a probability amplitude and $\hbar = h/(2\pi)$ is the reduced Planck constant. The solutions are $Ce^{\chi x} + De^{-\chi x}$, not an identically zero wave. Matching across a finite barrier can leave a transmitted wave. This is tunnelling, not temporary borrowing of energy. The actual D–T problem needs its varying Coulomb field and coupled nuclear channels.^[14]

3. From a possible reaction to a probability

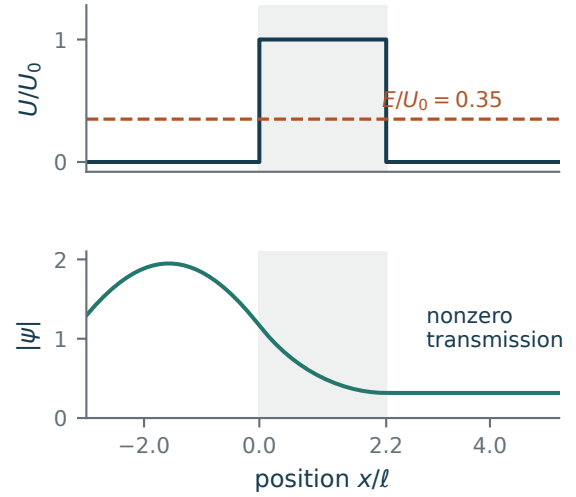
The symbol $D+T \rightarrow \alpha+n$ identifies an outcome. To predict its frequency, we need a quantum amplitude. A quantum state can be represented by a vector; time evolution changes that vector according to the interactions. For scattering, an operator S connects specified incoming states to outgoing states. In a simplified orthonormal wave-packet basis, write

$$S|i\rangle = \sum_f a_f |f\rangle, \quad a_f = \langle f|S|i\rangle, \quad P_f = |a_f|^2.$$

$|i\rangle$ denotes the prepared incoming state, $|f\rangle$ a resolved outgoing state, and a_f its complex coefficient. Orthogonality gives $\langle f|g\rangle = 0$ for distinct basis states and 1 when $f = g$; probability conservation then

A Binding energy across nuclei

AME2020 measured entries (no # estimates).
Line: largest tabulated B/A at each A.

B An approach barrier

Toy model: $\ell = \hbar/\sqrt{2\mu U_0}$.
Not a D-T cross-section calculation.

Figure 4: Different horizontal axes, different questions. (A) AME2020 entries with measured binding values and $A \leq 260$; estimated values marked # are excluded. The line is an upper envelope of the discrete entries, not a reaction path. Arrows indicate broad possible energy-releasing trends, not that every channel is favorable. (B) An exact stationary solution of the stated constant-barrier toy model: $E/U_0 = 0.35$, width 2.2ℓ , with $\ell = \hbar/\sqrt{2\mu U_0}$. Matching ψ and its derivative gives nonzero transmission; the lower curve is amplitude magnitude, not an energy curve. Original calculation; sources 11 and 14.

reads $\sum_f |a_f|^2 = 1$. Actual continuum momenta require integrals or finite detector bins rather than a probability for one exact momentum.^[15]

An amplitude includes both magnitude and phase. Contributions to the same indistinguishable outcome add *before* squaring; interference can therefore increase or suppress that outcome. We do not select nuclear products by counting nucleons alone.^[15]

How would we measure this? Send a narrow-energy deuteron beam toward a thin target containing N_T tritons. Let Φ_D be the incident number flux, in $\text{m}^{-2}\text{s}^{-1}$, and let $\dot{N}_{\text{fus}}$ count fusion events per second. In the dilute, independent-target limit, the operational definition is

$$\sigma_{DT}(E) = \frac{\dot{N}_{\text{fus}}}{\Phi_D N_T}. \quad (5)$$

Its units are area. It is the reaction's *effective* area at relative collision energy E , not the literal cross-sectional area of a little hard nucleus. The thin-target assumption avoids appreciable beam depletion and energy loss. Connecting an S -matrix element to this area requires incoming flux normalization, outgoing phase-space integration, and appropriate spin sums or averages; one cannot simply set $\sigma = |a|^2$.^[13]

Barrier penetration is only part of the amplitude. D-T fusion is enhanced by a resonance, a temporarily correlated five-nucleon configuration with the quantum numbers of unbound helium-5. A microscopic calculation by Hupin and colleagues treats incoming $D + T$ and outgoing $\alpha + n$ channels with nuclear interactions from effective field theory. Its high-accuracy comparison includes a small phenomenological resonance adjustment. This gives a concrete bridge from a microscopic nuclear model to measured reaction probabilities, with the model's assumptions still visible.^[16]

Where do path integrals enter? They provide another formulation of amplitudes: sum contributions from alternative histories with phases set by action divided by $\hbar$. Feynman diagrams organize particular expansions; they are not little movies of a measured trajectory. We cite this foundation, but do not claim to evaluate a nonperturbative QCD path integral here.^[17]

4. From one encounter to hot fuel

A reactor needs many reactions, not merely a possible one. Let n_D and n_T denote numbers of deuterons and tritons per unit volume. First imagine all encounters have relative speed v . In a short time dt , one deuteron samples an effective reaction volume $\sigma_{DT}v dt$. That volume contains, on average, $n_T\sigma_{DT}v dt$ tritons. Multiplying by n_D deuterons per unit volume gives the event rate density $\mathcal{R} = n_D n_T \sigma_{DT} v$.

Real particles do not all move at the same speed. Let $g(v) dv$ be the probability that a randomly selected D–T pair has relative speed in $[v, v + dv]$, with $\int_0^\infty g(v) dv = 1$. Averaging the preceding argument gives

$$\langle \sigma v \rangle = \int_0^\infty \sigma_{DT}(v) v g(v) dv, \quad \mathcal{R} = n_D n_T \langle \sigma v \rangle. \quad (6)$$

The brackets mean an average, not multiplication of separately averaged factors. Here $\sigma_{DT}(v)$ is the same cross section, indexed by relative speed rather than collision energy. For unlike species there is no factor 1/2: each D–T pair is counted once. This is a local, dilute, binary-collision model with uncorrelated incoming particles; it is not a formula for an arbitrary strongly correlated state. Dimensional checking gives $[\langle \sigma v \rangle] = \text{m}^3 \text{s}^{-1}$ and $[\mathcal{R}] = \text{m}^{-3} \text{s}^{-1}$. Multiplying by the energy per event gives the local *nuclear* power density, $\mathcal{P}_{\text{fus}} = Q\mathcal{R}$, not net electrical output.

For near-thermal fuel, ion temperature sets the velocity distribution. We keep temperature in kelvin, T_i , distinct from its energy scale, $\Theta_i = k_B T_i$, where k_B is Boltzmann’s constant. For example,

$$\Theta_i = 10 \text{ keV} = 10^4 \text{ eV} \quad \Rightarrow \quad T_i = \frac{10^4}{8.617333262 \times 10^{-5}} \text{ K} \simeq 1.16 \times 10^8 \text{ K}.^{[12]}$$

This is a scale illustration, not a temperature optimum derived from the Coulomb barrier. Evaluated cross sections, energy distributions, and losses must be combined to choose an operating range. The familiar multi-keV D–T regime will be justified by those calculations, not assumed to follow from “hotter is better.”^[18]

In the hot-fuel regime used for magnetic fusion, electrons no longer remain bound to individual nuclei: the fuel is an ionized plasma of positive nuclei and negative electrons. It is not a quark–gluon plasma; nucleons remain confined composite particles.^[1] Positive and negative charge can balance overall even though each constituent responds to electromagnetic fields.

The first machine question How can we give the fuel enough energetic encounters, and retain that energy long enough, for fusion power to compete with losses? In the magnetic-confinement route pursued here, sustained contact between the hot core and material walls would rapidly remove energy and add impurities. We need separation of core fuel from material surfaces. We have not yet earned a torus or a coil: next we ask whether electromagnetic forces can provide the required control.

The charged alpha particle and neutral neutron will later demand different energy-handling strategies. Microscopic color confinement and macroscopic plasma confinement are different problems; the latter does not follow from the former by analogy.

Open sources and the claims they support

These notes make the boundary between derivation and imported physics explicit. All links point to freely accessible material. The body develops its own bookkeeping, energy cancellation, recoil estimate, and rate-counting argument; the underlying physical laws and numerical inputs are not mathematical axioms proved by this thesis.

[1] Constituents and the meaning of quark–gluon plasma.

US Department of Energy, “[DOE Explains: Quarks and Gluons](#).” Introductory orientation for quark content, gluons, and the distinction between intact nucleons and deconfined matter. The electron-cloud description does not assume classical orbital trajectories.

[2] Color charge and gluon self-interactions.

Particle Data Group (2025), “[Quantum Chromodynamics](#),” §9.1, especially Eqs. (9.1)–(9.2) and the discussion of three- and four-gluon vertices on p. 2. These specify the microscopic theory; we do not derive its Lagrangian or Feynman rules in the opening.

[3] Nucleon mass from dynamical QCD.

S. Dürr et al., “[Ab-initio Determination of Light Hadron Masses](#),” *Science* **322**, 1224–1227 (2008), open manuscript. Consult the mass-spectrum results and discussion of continuum and physical-mass extrapolations. Quark masses and the overall physical scale are inputs; this does not assign all mass to a single geometric flux-tube component.

[4] Asymptotic freedom.

D. J. Gross, “[The Discovery of Asymptotic Freedom and the Emergence of QCD](#),” Nobel lecture (2004), especially “Implications of Asymptotic Freedom.” For the precise running-coupling equation, see source 2, §9.1.1, Eq. (9.3). A weak-coupling result at short distance is not by itself a long-distance confinement proof.

[5] Flux tubes, a static potential, and string breaking.

G. S. Bali et al. (SESAM), “[Observation of String Breaking in QCD](#),” *Physical Review D* **71**, 114513 (2005). The system has static quark sources and dynamical light quarks. The linear potential is an intermediate-regime model; the final states after breaking are color neutral. It is not a rigid-string model of a physical proton.

[6] What the unsolved mathematical problem actually asks.

A. Jaffe and E. Witten, “[Quantum Yang–Mills Theory](#),” §4 (printed p. 6) for existence and the spectral-gap statement; §5 distinguishes extensions including confinement. The full statement treats any compact simple gauge group on four-dimensional spacetime with specified axiomatic properties. The [Clay problem-status page](#) still lists the missing proof (checked 11 September 2026). This is not the claim that gluons acquire an ordinary particle mass.

[7] A scale-dependent-coupling analogue outside nuclei.

P. Coleman, “[Phil Anderson’s Magnetic Ideas in Science](#)” (2015), §3.5, printed p. 10. The Kondo example concerns impurity-spin screening in a metal, not a nuclear force or a proposal for plasma confinement.

[8] The QCD-to-nuclear-force bridge.

R. Machleidt and D. R. Entem, “[Chiral Effective Field Theory and Nuclear Forces](#),” *Physics Reports* **503**, 1–75 (2011). See the introductory discussion and sections on the chiral expansion, pion exchange, contact interactions, and many-nucleon forces. This is a systematic effective description with fitted inputs and truncation errors, not escaping color flux.

Open sources, continued

[9] Nuclear notation and the binding-energy trend.

P. Cappellaro, [MIT 22.02, Chapter 1: Introduction to Nuclear Physics](#) (2012), §1.1.1 and §1.2. The liquid-drop explanation is qualitative here; no fitted mass formula is used to predict light-nucleus binding energies or the exact peak location.

[10] The deliberately short weak-interaction detour.

L. E. Marcucci, R. Schiavilla, and M. Viviani, “[The Proton–Proton Weak Capture in Chiral Effective Field Theory](#)” (2013). The solar reaction is not free proton decay, nor the only physical route by which deuterium can form.

[11] Numerical binding-energy inputs.

Atomic Mass Evaluation 2020, [mass_1.mas20](#) from the IAEA Atomic Mass Data Center. Rows $(Z, A) = (1, 2), (1, 3), (2, 4), (26, 56), (28, 62)$ provide the relevant values; the binding column is *keV per nucleon*, so multiply by A and divide by 1000 for total MeV. Atomic-mass conventions require electronic-binding corrections for high-precision bare-nucleus energies; they are immaterial to the 0.001 MeV inputs used here.

[12] Units and temperature conversion.

NIST, [SI Brochure, §2: The International System of Units](#). Exact defining values: $e = 1.602176634 \times 10^{-19}$ C and $k_B = 1.380649 \times 10^{-23}$ J K^{−1}. Their quotient gives $k_B = 8.617333262 \dots \times 10^{-5}$ eV K^{−1}. Together with NIST’s [CODATA table entry for vacuum electric permittivity](#), these give $e^2/(4\pi\epsilon_0) \simeq 1.440$ MeV fm.

[13] Two-body kinematics and the meaning of cross section.

Particle Data Group (2025), “[Kinematics](#),” §49.3 (invariant amplitudes), §49.4.2 (two-body final states), and §49.5 (cross sections). The main-text recoil result is a leading nonrelativistic estimate, not an exact relativistic energy partition.

[14] The quantum law used in the barrier illustration.

R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. III, Chapter 16, especially the Schrödinger-equation discussion. Our constant-barrier ODE illustrates penetration only; it is not used as a numerical D–T cross-section model.

[15] Amplitudes and interference.

Feynman, Leighton, and Sands, Vol. III, Chapter 3: [Probability Amplitudes](#), especially the rules for combining amplitudes. The main text uses a normalized wave-packet basis to avoid assigning a finite probability to one exactly specified continuum momentum.

[16] A real microscopic D–T calculation and its limits.

G. Hupin, S. Quaglioni, and P. Navrátil, “[Ab Initio Predictions for Polarized Deuterium–Tritium Thermonuclear Fusion](#),” *Nature Communications* **10**, 351 (2019). See Results, Fig. 1, and Methods: the “NCSMC-pheno” result includes a -5 keV resonance centroid correction. No polarized-fuel enhancement is assumed in our opening.

[17] Action and the sum over histories.

Feynman, Leighton, and Sands, Vol. II, Chapter 19: [The Principle of Least Action](#), concluding quantum discussion. This supplies an accessible conceptual entry, not a construction of the QCD functional measure.

[18] Later engineering connection.

J. P. Freidberg, F. J. Mangiarotti, and J. Minervini, “[Designing a Tokamak Fusion Reactor—How Does Plasma Physics Fit In?](#)” (2015), MIT report PSFC/JA-15-21. This is the subsequent design target, not the source of a confinement proof or a universal temperature threshold.